UTC Chemical Engineering • Student Lab Guide

CO_2 Capture Challenge

Part 1 — Setup & CO_2 Capture

Goal

Generate CO_2, expose NaOH-soaked foam to the gas, and observe signs of capture.

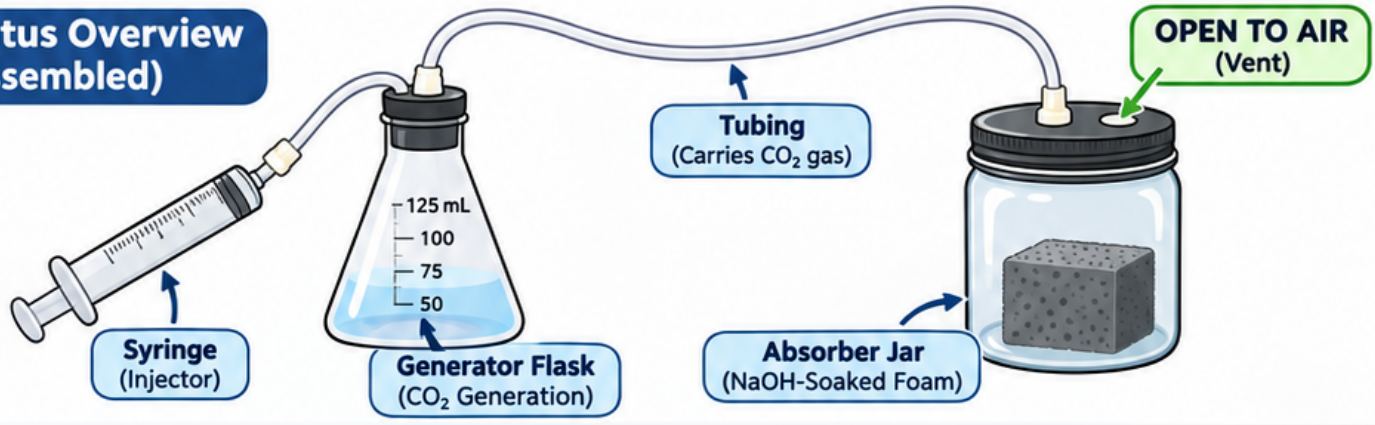
Safety

- Wear goggles, gloves, lab coat, closed-toe shoes
- NaOH is corrosive
- Leave the absorber vent open
- Never force the syringe
- Tell the instructor if a tube is blocked or leaking

Materials per group

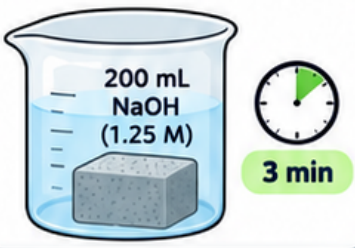
- 1 foam cube (~4 × 4 × 3 cm)
- 200 mL 1.25 M NaOH
- 1.00 g baking soda
- 15 mL 5% vinegar
- 125 mL generator flask
- Absorber jar with tubing + open vent
- Shared syringe / injector

Apparatus Overview (Assembled)



Procedure: 7 Easy Steps

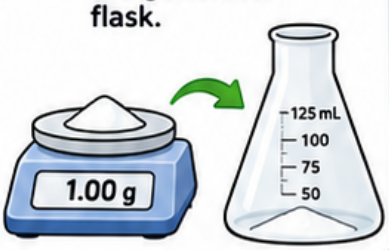
1 Soak **ONE foam cube in 200 mL NaOH** for **3 min.**



2 Lift out the cube, let it drain, and place **only** the foam in the absorber jar. **Do NOT** pour the soaking solution into the jar.



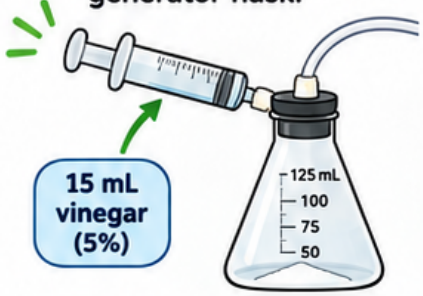
3 Place **1.00 g** baking soda into the generator flask.



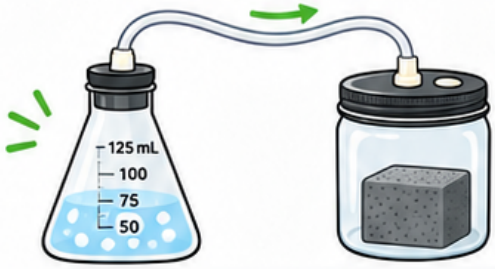
4 Instructor checks the setup: secure tubing, open vent, **safe connections.**



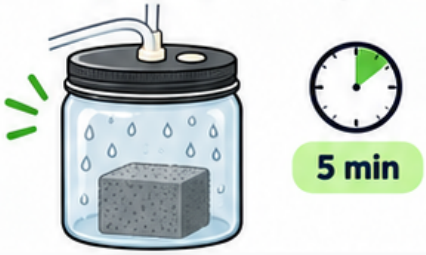
5 Slowly inject **15 mL** vinegar into the generator flask.



6 Watch the bubbling. CO_2 travels through the tubing into the jar.



7 Let the foam sit in the CO_2 stream for **5 min.** Look for fogging / condensation in the jar.



One cube per jar.

Do not disconnect tubing during gas generation.

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Part 2 — Precipitation & Filtration

What happens next?

Move the CO_2-exposed foam into $CaCl_2$ solution, form a white precipitate, and collect it by vacuum filtration.

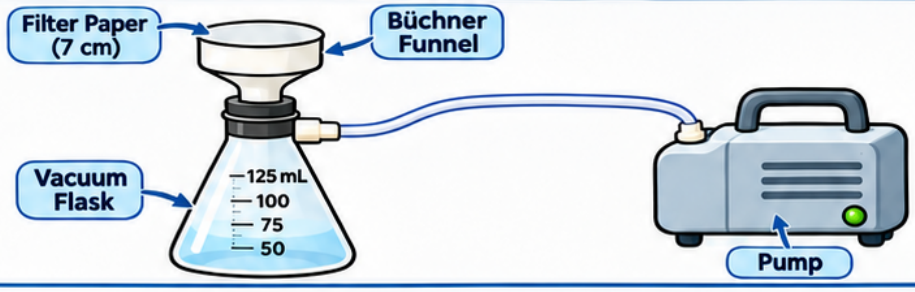
Safety

- Handle wet foam with tools
- Keep vacuum setup instructor-approved
- Do not pour reaction liquids down the drain
- Report broken glass or loose tubing

Key amounts

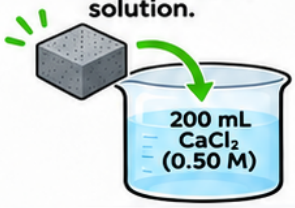
- 200 mL 0.50 M $CaCl_2$
- 5 min precipitation time
- 7 cm filter paper
- Record dry paper mass first
- Final result = **APPARENT WET** recovery

Apparatus Overview (Filtration Setup)

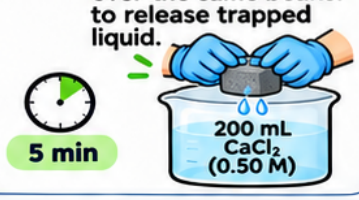


Procedure: 7 Easy Steps

1 Transfer the exposed foam cube into 200 mL $CaCl_2$ solution.



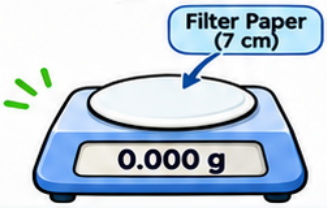
2 Let it sit for 5 min, then gently squeeze / compress the foam over the same beaker to release trapped liquid.



3 Look for white cloudiness or precipitate in the $CaCl_2$ beaker.



4 Weigh the dry 7 cm filter paper before it gets wet.



5 Assemble Büchner funnel + vacuum flask + pump with instructor help.



6 Seat the filter paper using a small amount of reaction liquid. No extra DI water is needed here. Then filter the rest of the mixture.



7 Weigh the damp filter paper + collected solid and calculate apparent WET recovery.



Record these observations:

- ☐ Did the jar fog?
- ☐ Did white precipitate appear?
- ☐ Dry filter mass (g): _____
- ☐ Damp filter + product mass (g): _____

Wet recovery is NOT the same as dry $CaCO_3$ yield.